

PERAK, Malaysia

WMO: 486200

Lat: 4.2211N

Lon: 100.7010E

Elev: 26

StdP: 14.68

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	72.3	73.1	69.5	108.4	80.4	70.7	113.2	79.9	11.3	85.2	9.9	85.7	1.4	70	0.307

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	13.9	93.1	80.3	91.8	80.1	91.4	80.0	82.4	89.9	81.8	89.1	81.3	88.5	6.3	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.6	158.9	87.4	79.7	154.1	86.2	79.1	151.4	85.6	46.4	89.4	45.6	90.0	45.1	89.1	88.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.9	9.5	8.3	70.0	94.3	1.6	1.2	68.9	95.2	68.0	95.9	67.1	96.6	65.9	97.4		
(5)			68.6	83.9	1.7	1.7	67.4	85.1	66.4	86.1	65.4	87.0	64.2	88.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	82.1	81.2	81.9	82.7	83.1	83.3	83.0	82.3	82.1	81.9	81.6	81.4	81.2
	DBStd	1.68	1.53	1.54	1.70	1.62	1.41	1.52	1.49	1.54	1.51	1.52	1.32	1.50
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD50	11733	969	894	1015	992	1033	989	1002	996	957	979	941	966
	CDD65	6258	504	474	550	542	568	539	537	531	507	514	491	501
	CDH74	63317	4742	4790	5779	5850	6308	5893	5622	5414	5029	4879	4475	4536
	CDH80	24384	1687	1898	2388	2401	2597	2459	2270	2117	1857	1712	1495	1503
Wind	WSAvg	2.6	2.4	2.7	2.8	2.8	2.6	2.6	2.6	2.7	2.7	2.8	2.4	2.3
Precipitation	PrecAvg	68.0	5.2	5.3	4.3	5.9	4.7	3.2	3.7	4.1	6.7	8.5	8.7	7.5
	PrecMax	82.8	10.1	11.7	14.2	11.6	9.9	7.4	10.0	10.1	12.2	22.2	14.8	13.9
	PrecMin	54.3	0.3	1.2	0.9	0.4	1.4	0.5	0.3	1.1	1.2	3.5	4.1	3.3
	PrecStd	7.4	2.6	3.1	2.8	2.6	2.5	1.7	2.0	2.1	2.6	3.3	3.1	2.9
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.6	93.1	93.6	93.5	93.4	93.4	93.0	93.0	92.2	91.7	90.9	91.2
		MCWB	79.4	79.5	80.6	81.1	80.8	80.2	79.7	79.2	79.2	79.4	80.4	80.0
	2%	DB	89.9	91.4	91.8	91.8	91.8	91.7	91.4	91.3	90.9	90.0	89.6	89.6
		MCWB	78.9	79.0	80.1	80.7	80.8	80.2	79.7	79.4	79.3	79.4	79.8	79.4
	5%	DB	88.9	89.9	91.0	91.0	91.1	91.1	90.0	89.9	89.6	89.2	88.2	88.2
		MCWB	78.8	78.8	79.8	80.5	80.7	80.1	79.5	79.2	79.2	79.3	79.7	79.2
	10%	DB	87.6	88.8	89.6	89.6	89.7	89.7	89.3	88.8	88.1	87.7	87.4	87.2
		MCWB	78.6	78.7	79.5	80.1	80.4	79.8	79.3	79.1	79.0	79.2	79.4	79.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	81.7	81.6	82.6	82.7	83.3	82.8	82.0	81.7	81.7	81.5	82.1	81.7
		MCDB	88.5	89.8	91.0	90.4	90.9	90.4	89.7	89.6	89.1	88.3	88.8	88.8
	2%	WB	80.4	80.6	81.5	81.8	82.3	81.8	81.1	80.7	80.6	80.5	80.9	80.6
		MCDB	87.3	88.3	89.4	89.3	89.3	89.1	88.5	88.2	87.7	87.5	87.3	87.3
	5%	WB	79.7	79.9	80.7	81.1	81.6	81.0	80.3	80.0	79.9	79.9	80.1	79.9
		MCDB	86.2	87.2	88.3	88.4	88.4	88.1	87.4	87.2	86.7	86.5	86.1	86.1
	10%	WB	79.1	79.2	80.0	80.5	80.9	80.3	79.7	79.4	79.3	79.2	79.5	79.2
		MCDB	85.2	86.1	87.1	87.5	87.5	87.2	86.7	86.4	85.8	85.5	85.3	85.0
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.9	14.7	14.0	13.9	14.4	14.5	14.2	13.7	13.1	12.7	12.9
		MCDBR	15.3	16.2	16.1	15.2	15.3	15.7	15.7	15.6	15.3	14.7	14.0	14.4
		MCWBR	6.3	6.3	6.0	5.8	5.9	6.1	6.0	6.0	5.8	5.8	5.8	5.8
	5% WB	MCDBR	14.0	15.0	14.9	14.5	14.1	14.6	14.7	14.6	14.1	13.7	13.2	13.3
MCWBR		6.2	6.2	6.1	5.9	6.1	6.2	6.1	6.0	5.9	5.9	5.9	5.9	
Clear-Sky Solar Irradiance	taub	0.470	0.514	0.515	0.501	0.479	0.507	0.501	0.509	0.513	0.534	0.475	0.461	
	taud	2.285	2.135	2.161	2.205	2.279	2.205	2.213	2.178	2.167	2.104	2.285	2.328	
	Ebn at Noon	271	262	261	259	259	248	251	254	258	253	268	272	
	Edh at Noon	43	51	50	47	43	45	45	48	49	52	43	41	
All-Sky Solar Radiation	RadAvg	1527	1737	1773	1756	1639	1626	1591	1619	1553	1512	1460	1411	
	RadStd	118	111	114	57	67	69	68	68	106	127	70	103	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		+0.79	+0.56	+0.47	+1.01	+0.85	+0.90	N/A	N/A	+292
(47) Regional (4 neighbors)		+0.70	+0.40	N/A	+0.94	+0.79	+0.67	N/A	N/A	+252

Nomenclature: See separate page